

# The Denominator Rule

*A percentage without its denominator is not a result. It is a rumour.*

## The Rule

**Never report a percentage without stating the denominator, and never choose the denominator after you have seen the answer.**

## The same data. Three answers. All arithmetically correct.

In the practice dataset there are **180 patients** registered for TB treatment. Of these, **126 were cured or completed treatment**. So what is the treatment success rate?

| Denominator you choose                           | n   | Success rate | What you are actually saying                                                  |
|--------------------------------------------------|-----|--------------|-------------------------------------------------------------------------------|
| A. Everyone registered                           | 180 | <b>70.0%</b> | Includes 8 patients still on treatment, whose outcome is not yet known        |
| B. Excluding those still on treatment            | 172 | <b>73.3%</b> | Fairer — but still includes 12 patients transferred out and never followed up |
| C. Only patients with a known, evaluated outcome | 160 | <b>78.8%</b> | The WHO cohort definition — but hides the 12 you lost track of                |

### Stop and look at that range

**70.0% to 78.8%.** Nearly nine percentage points, from the same 180 patients, with no arithmetic error anywhere. If your national target is 75%, you have just 'passed' or 'failed' depending on a choice you made in a spreadsheet — and if you do not state which choice you made, no reader can tell.

## What to do instead

- 1. Decide your denominator in your protocol**, before you see the results. Write it down. This is the whole reason we make you write an analysis plan.
- 2. Use the standard definition** for your disease programme where one exists. For TB outcomes, use the WHO cohort definition and say so.
- 3. Report n and N every single time.** Write '126/160 (78.8%)', never '78.8%' alone.
- 4. Account for everybody.** The patients you excluded do not disappear — they are a finding. '12 patients (6.7%) were transferred out and their outcome could not be traced' is often the most operationally useful sentence in the whole paper.
- 5. Show a flow diagram** if numbers were excluded at several stages, so the reader can follow every patient from N registered to n analysed.

## The small-denominator trap

In the same dataset, treatment success among HIV-positive patients is **13/13 = 100%**.

### A 100% success rate that means almost nothing

Do **not** write 'HIV-positive patients had a 100% treatment success rate'. With a denominator of 13, one additional death would have moved that figure to 92.9%. Percentages calculated on fewer than about 20–30 people are

extremely unstable. **Below 20, report the raw counts and skip the percentage altogether.**

## Denominator checklist — run this before every table you publish

- Does every percentage in my table have its n and N shown?
- Do my percentages add to 100% (or do I explain why they do not)?
- Is the denominator the same in every row, or does it change? If it changes, is that obvious to the reader?
- Have I stated how I handled missing values — excluded, or counted as a category?
- Is any denominator below 30? If so, have I reported counts instead of percentages?
- Would a reader who has never seen my dataset be able to reconstruct my table from my methods section?

### Liafuan importante • Key words in Tetum

|                     |             |
|---------------------|-------------|
| Denominator         | Denominadór |
| Percentage          | Pursentu    |
| Total               | Totál       |
| Result              | Rezultadu   |
| Success             | Susesu      |
| Missing / not there | La iha      |
| Patient             | Pasiente    |